Report R17 — what the exponential-exponential attractors actually look like: dense, compact, and borrowed

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-07

Abstract

Four classes in R9's F1 right panel have both coordinates above 1: exponentially many monitored attractors, each of which also grows exponentially. This report asks whether those attractors are simply long cycles — all a V-free rule can offer, its map being a function — or something more connected. Neither alternative survives contact with the graphs. Of the 22 945 terminal SCCs the twelve rules have between them at $N = 16$, not one with more than a single state is a cycle. They are **not cycles** and they are **not extended**: every one has a self-loop on every state, an edge density of order 1/2, and a directed diameter of 2 at every size measured but one (a single 3), up to 2584 states and 2.9×10^6 edges. The state count grows like φ^N; the graph distance does not grow at all. The extreme case is the plastic-number class (28, 29, 70, 71, 157, 199), whose every terminal SCC, for all six rules at every N, is a *complete* digraph K_n with all n^2 edges including loops — a uniform transition matrix, second eigenvalue exactly 0, memoryless after one period. Its size is $2^{\lfloor (N+1)/3 \rfloor}$ exactly, which supplies a derivation for the base $2^{1/3}$ that R9 fitted. Underneath all four classes sits one fact: there are not four structures but **two**. The twelve rules are two coherent reflection pairs, W_{108}/W_{201} and W_{156}/W_{198}, plus their eight dissipative children, and for every child the terminal SCCs are Krylov sectors of the parent with the same vertex set *and* the same induced edges. The reset never fires on its own attractor; it is a selection rule, and which sectors it selects is the entire difference between the classes. On the graph-theoretic measures: the graphs are not too small, but clustering and connectivity are uninformative *because of the density* — both sit at their generic values — and we report them with their nulls rather than as findings.

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1 The question and the twelve rules

R9's F1 places every rule at $(a_{\text{att}}, b_{\text{att}})$: the growth base of the number of terminal SCCs against that of the largest one. Most of the map lies on $b_{\text{att}} = 1$ — many attractors, each of bounded size — and R15 accounted for the column at $a_{\text{att}} = 1$. Four classes sit off both axes:

class	rules	tuples	a_{att}	b_{att}	family
1	W_{108}, W_{201}	`IIIV`, `VIII`	$\rho^2 = 1.7549$	$\varphi = 1.6180$	unitary
2	W_{156}, W_{198}	`IIVI`, `IVII`	$\varphi = 1.6180$	$4^{1/5} = 1.3195$	unitary
3	W_{73}, W_{109}	`VIID`, `EIIV`	$\psi = 1.4656$	$\varphi = 1.6180$	V+reset
4	$W_{28}, W_{29}, W_{70}, W_{71}, W_{157}, W_{199}$	—	$\rho = 1.3247$	$2^{1/3} = 1.2599$	V+reset

with φ the golden ratio ($x^2 = x + 1$), $\psi = 1.465571$ the supergolden ratio ($x^3 = x^2 + 1$) and $\rho = 1.324718$ the plastic number ($x^3 = x + 1$). Class 1's constant is not a fifth number: $\rho^2 = 1.754878$ exactly, a coincidence that turns out to be structural rather than numerical (§3).

The question put to this report was whether these attractors are simply cycles, or have a more connected and extended structure distinct from the V-free circuits — and whether graph-theoretic measures can say anything at these sizes, or whether the objects are still too small.

Why "cycle" is the natural null. A V-free rule has a *deterministic* update: out-degree 1 at every state, so its transition graph is a functional graph and a terminal SCC can only be a cycle. That is not an observation, it is a definition, and we check the premise rather than assume it: all 81 V-free rules have out-degree 1 everywhere. What they do not have is length. At $N = 12$, 75 of the 81 have nothing but fixed points, two have cycles of length 2, two of length 3, and the longest cycle anywhere in the V-free half of rule space is 63 (rules 90 and 165, the two whose series R9 classifies as irregular).

V-free control at $N = 12$	value
rules	81
out-degree 1 everywhere	all
longest cycle anywhere	63
rules with fixed points only	75
cycle-length histogram	1: 75, 2: 2, 3: 2, 63: 2

2 None of them is a cycle

2.1 The measurements

Table 2.1 gives the largest terminal SCC of one representative per class at $N = 16$.

class $(a_{\text{att}}, b_{\text{att}})$	rule	tuple	N	n	m	m/n^2	diam	$\bar{d}$	recip.	$\lvert\lambda_2\rvert$	complete
ρ^2, φ	W_{201}	`VIII`	16	2584	2875072	0.431	2	1.570	0.319	0.2993	no
φ, $4^{1/5}$	W_{198}	`IVII`	16	108	6144	0.527	2	1.478	0.541	0.3333	no
ψ, φ	W_{109}	`EIIV`	16	987	467008	0.479	2	1.521	0.369	0.2986	no
ρ, $2^{1/3}$	W_{199}	`EVII`	16	32	1024	1.000	1	1.000	0.969	0.0000	yes

A cycle on n nodes has $m = n$ edges, density $1/n$, and diameter $n - 1$. These have m of order $n^2/2$, density that never falls below 0.43, and diameter 2 — with one exception across all twenty measurements, class 2 at $N = 10$, where it is 3. W_{201} at $N = 16$ is a graph on 2584 states with 2 875 072 edges: every state has a self-loop, the mean out-degree is 1112, and any state reaches any other in at most two brick-wall periods.

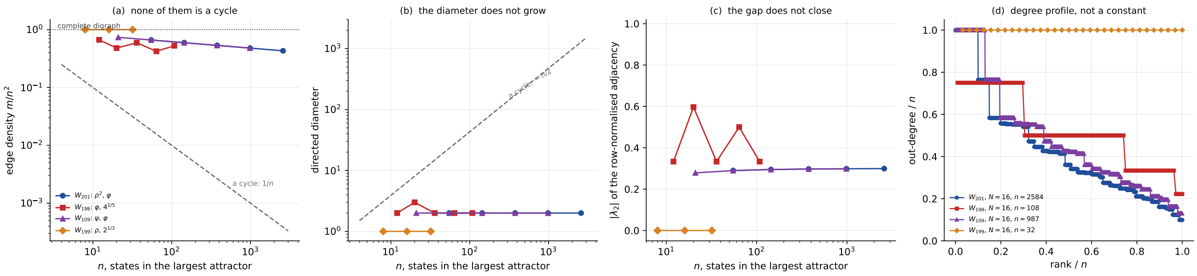

Figure 1: **The four classes against the cycle they are not.** (a) Edge density against attractor size, with the $1/n$ a cycle would give as a dashed reference and 1 (a complete digraph) as a dotted one; the plastic class sits *on* the upper line. (b) Directed diameter, against the $\sim n/2$ of a cycle: it is 2 everywhere and 1 for the complete class, over two decades in n. (c) The second-largest eigenvalue modulus of the row-normalised adjacency — the mixing rate of the monitored walk inside the attractor. It does not drift towards 1 as the attractor grows; the complete class sits at exactly 0. (d) The out-degree profile, sorted and normalised by n: flat at 1 for the complete class, a structured staircase for the others, and nowhere the constant $1/n$ of a cycle.

2.2 The plastic class is a complete digraph, exactly

Proposition 1. *For each of W_{28}, W_{29}, W_{70}, W_{71}, W_{157}, W_{199} at open boundaries,* every *terminal SCC — not merely the largest — is the complete digraph on its states, all n^2 edges present including every self-loop. Verified exhaustively for $N = 8, 10, 12, 14, 16$.*

Two consequences follow immediately and neither needs a fit. First, the transition matrix restricted to such an attractor is uniform, so all its non-Perron eigenvalues are 0: the monitored dynamics inside the attractor is *memoryless after a single period*, and the diameter is 1. Second, the attractor sizes are exactly powers of two,

$$D^{\text{att}}_{\max}(N) \;=\; 2^{\lfloor (N+1)/3 \rfloor}, \qquad N = 6, \ldots, 16: \quad 4, 4, 8, 8, 8, 16, 16, 16, 32, 32, 32, \tag{1}$$

which gives $b_{\text{att}} = 2^{1/3}$ *exactly* rather than as R9's fitted value. The attractor counts run $11, 15, 20, 27, 36, 48, 64, 85, 113, 150, 199$ for W_{28} and $5, 7, 9, 12, 16, 21, 28, 37, 49, 65, 86$ for W_{199}; both ratios settle at ρ^2 per two sites.

rule	tuple	N	attractors	fixed pts	> 1 state	complete	cycles	other	all complete
W_{108}	IIIV	16	10252	3025	7227	2736	0	4491	no
W_{201}	VIII	16	5842	1156	4686	1547	0	3139	no
W_{156}	IIVI	16	2584	9	2575	190	0	2385	no
W_{198}	IVII	16	2584	9	2575	190	0	2385	no
W_{73}	VIID	16	406	16	390	57	0	333	no
W_{109}	EIIV	16	535	37	498	90	0	408	no
W_{28}	IIVD	16	199	9	190	190	0	0	yes
W_{29}	EIVD	16	86	1	85	85	0	0	yes
W_{70}	IVID	16	199	9	190	190	0	0	yes
W_{71}	EVID	16	86	1	85	85	0	0	yes
W_{157}	EIVI	16	86	1	85	85	0	0	yes
W_{199}	EVII	16	86	1	85	85	0	0	yes

That table also settles the headline question far more strongly than the largest attractor alone can. It classifies *every* terminal SCC of all twelve rules at $N = 16$ — 22 945 of them — and the "cycles" column is **0** in every row. Not one attractor with more than a single state, in any of the four classes, is a cycle. (Fixed points are broken out separately because a lone state with a self-loop is trivially both a cycle and a complete digraph, and counting it as either would be bookkeeping dressed as evidence.) The complete count also closes the parent argument numerically: W_{156} and W_{198} have 190 complete non-trivial sectors at $N = 16$, and W_{28} and W_{70} have exactly 190 non-trivial attractors, all of them complete.

2.3 The spectral gap does not close

The informative dynamical statement is not the diameter — a dense graph has a small diameter for cheap reasons — but the mixing rate. For the φ-classes $|\lambda_2|$ of the row-normalised adjacency runs

$$0.2786,\ 0.2899,\ 0.2949,\ 0.2973,\ 0.2986,\ 0.2993$$

as the attractor grows through $21, 55, 144, 377, 987, 2584$ states: flat to three digits over two decades in n, and creeping *up* rather than towards 1. The mixing time inside the attractor is $O(1)$ while the attractor itself is $\Theta(\varphi^N)$. For the plastic class the gap is maximal — $|\lambda_2| = 0$ exactly. Class 2 is the one that does not settle to a single number: W_{156} reads $0.5, 0.597, 0.5, 0.5, 0.5$ and W_{198} reads $0.333, 0.597, 0.333, 0.5, 0.333$ over $N = 8 \ldots 16$, oscillating inside $[1/3, 3/5]$ with N mod 4. It is worth noting that the two differ at all: they are a reflection pair with identical sizes, densities and reciprocities, and their out- and in-degree distributions are exact mirrors of each other — but P is row-normalised, which is not a reflection-covariant operation, so the mixing rate of the forward walk need not agree. In none of the four classes does the gap close as the system grows, which is the point.

2.4 What grows, and what does not

The density does fall, and slowly: $0.735, 0.661, 0.594, 0.534, 0.479, 0.431$ over $N = 8 \ldots 16$, a factor 0.900 per two sites, i.e. $\sim 0.9487^N$. So the graphs thin asymptotically, while the mean out-degree grows like 1.534^N and the edge count like 2.481^N. We report those two bases as fitted; a search over minimal polynomials of degree ≤ 3 with small integer coefficients does not identify either, unlike every base R9 reports for these rules, and we leave that open rather than assert an algebraic form we cannot exhibit.

3 There are two structures here, not four

Every one of the twelve rules has a coherent parent obtained by replacing each reset symbol with the identity, and there are only two parents:

$$W_{73} \to W_{201}, \quad W_{109} \to W_{108}; \qquad W_{28}, W_{29}, W_{157} \to W_{156}, \quad W_{70}, W_{71}, W_{199} \to W_{198}.$$

Classes 1 and 2 *are* those parents. The content of this section is that classes 3 and 4 are as well.

Proposition 2. *For each of the eight dissipative rules and for $N = 8, 10, 12$, every terminal SCC of the child is a Krylov sector of its coherent parent — the same set of states — and the graph the child induces on it has* the same edge set *as the parent's. The reset writes nothing anywhere on its own attractors.*

child		parent		N	attractors	sectors	same sets	same edges	parent complete	child vs those
W_{73}	`VIID`	W_{201}	`VIII`	12	88	616	yes	yes	335	—
W_{109}	`EIIV`	W_{108}	`IIIV`	12	116	1081	yes	yes	715	—
W_{28}	`IIVD`	W_{156}	`IIVI`	12	64	377	yes	yes	64	all
W_{29}	`EIVD`	W_{156}	`IIVI`	12	28	377	yes	yes	64	subset
W_{70}	`IVID`	W_{198}	`IVII`	12	64	377	yes	yes	64	all
W_{71}	`EVID`	W_{198}	`IVII`	12	28	377	yes	yes	64	subset
W_{157}	`EIVI`	W_{156}	`IIVI`	12	28	377	yes	yes	64	subset
W_{199}	`EVII`	W_{198}	`IVII`	12	28	377	yes	yes	64	subset

Set equality alone would be the weak version of this claim: a child could inherit the states and still have different edges, since a reset that fires merges branches. It does not. So the dissipation here creates no new structure at all; it acts purely as a **selection rule** on the parent's sector list, and the whole difference between the four classes is which sectors survive:

class	what it keeps	a_{att}	b_{att}
108, 201	every sector	ρ^2	φ
73, 109	a ψ-sized subfamily, including the largest	ψ	φ
156, 198	every sector	φ	$4^{1/5}$
$28, 29, 70, 71, 157, 199$	the sectors that are already complete digraphs	ρ	$2^{1/3}$

The last row is exact for the pure-`D` children: W_{28}'s attractors are *precisely* W_{156}'s complete sectors, 36, 64 and 113 of them at $N = 10, 12, 14$, matching its attractor count series term for term. The children carrying an `E` keep a strict subset — W_{199} retains 16, 28, 49 of W_{198}'s 36, 64, 113 — and both still count at ρ per site. Recording that difference rather than smoothing it is the honest version: "the plastic class is the complete-sector subfamily" is exactly right for two of the six rules and one containment short for the other four.

This also explains the coincidence noted in §1. $b_{\text{att}} = \varphi$ is shared by classes 1 and 3 because the largest surviving sector is the same object; the two classes differ only in a_{att}, i.e. in how many sectors the reset spares, and ρ^2 against ψ is a counting difference within one family of graphs rather than two different kinds of dynamics.

4 Which graph measures earn their place

The question included whether the objects are still too small for graph-theoretic measures. They are not: n reaches 2584 with 2.9M edges, which is ample for degree distributions, distances, connectivity and spectra. But two standard measures are uninformative here, and the reason is the density rather than the size. We would rather say so than present a saturated number as a result.

Clustering says nothing. At density p, a random digraph already has transitivity $\approx p$. W_{201} at $N = 16$ measures 0.805 against a density-matched null of 0.723: a real but small excess on a scale that is nearly full. The module therefore never reports transitivity without its null beside it, and a regression test enforces that.

Connectivity says nothing either. In every case small enough to compute it exactly ($n \leq 200$, since it is n max-flow solves and costs two minutes already at $n = 377$), the vertex and edge connectivities are equal to each other *and* to the minimum degree: $\kappa_v = \kappa_e = \delta = 37$ for W_{108} at $N = 10$, 7 for W_{156}, 7 for W_{199}. That is the generic value for a dense graph; it describes the sparsest vertex, not the structure.

What does discriminate. Four things, all of which appear in Fig. 1: the density and its scaling, the degree profile (a structured staircase taking $O(\log n)$ distinct values, with the maximum out-degree equal to n itself — some states reach the entire attractor in one period), the reciprocity, which falls steadily from 0.605 at $n = 21$ to 0.319 at $n = 2584$ and so certifies that these graphs are genuinely directed rather than symmetrised walks, and the spectral gap.

5 Answering the question as asked

1. **Are they simply cycles?** No, and not marginally. Of the 22 945 terminal SCCs the twelve rules have between them at $N = 16$, *none* with more than one state is a cycle. Every one carries a self-loop on every state, has density of order 1/2 or exactly 1, and has diameter 2 or 1 (once 3). A cycle has density $1/n$ and diameter $n - 1$.

2. **Are they a more connected and extended structure?** More connected, yes — dramatically. *Extended*, no: that is the surprise. The number of states grows exponentially and the graph diameter stays at 2. These are compact, densely wired objects, and the monitored walk inside one mixes in $O(1)$ periods no matter how large it is.

3. **Distinct from the V-free circuits?** Distinct in kind, not in degree. A V-free rule cannot produce anything but a cycle, because its map is a function; the branching that makes these graphs two-dimensional is exactly the Hadamard. And the V-free circuits are not even long: 75 of 81 rules have only fixed points.

4. **Are they too small for graph measures?** No — but clustering and connectivity are uninformative on graphs this dense, and we report both against their null values rather than as findings. Density scaling, degree profile, reciprocity and the spectral gap are the measures that separate these graphs from a random one.

5. **Unlooked-for, and the main result.** The four classes are two coherent structures. All eight dissipative rules inherit their parent's sectors whole, edges included, and differ only in which sectors the reset spares — which for the plastic class is exactly the complete ones, and which turns $b_{\text{att}} = 2^{1/3}$ from a fitted constant into $2^{\lfloor (N+1)/3 \rfloor}$.

6 Reproduction

```
python -m qca_fragmentation.scaling.attractor_topology --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_attractor_topology.py -q
```

Everything in this report comes from `analytics/attractor_topology_obc0.json`. Two implementation notes, because both changed what was feasible to measure. All-pairs distances are computed by boolean matrix powers rather than by BFS from every node: networkx's $O(nm)$ pass is $\sim 10^{10}$ Python steps at $n = 2584$, $m = 2.9$M, whereas the diameter being 2 means the matrix loop runs twice, each step one BLAS product on a 26 MB `float32` array — a trade that is only right *because* the answer is small, and a long-diameter graph would want BFS. Transitivity is likewise taken as $\mathrm{tr}(B^3)/\sum_i d_i(d_i - 1)$ rather than by enumerating triangles. Connectivity has no such shortcut and is the one measure that is capped by size rather than computed everywhere.